
Solution to Exercise Sheet 3:
Constraint Satisfaction Problems

Foundations of Neuro-Symbolic AI
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Task 1:

- (i) We enumerate each row by an index $r \in [n]$ and each cell by an index $c \in [n]$. We then define for each cell representing tuple (r, c) a variable X_{rc} with $m_{rc} = 2$, indicating whether the cell contains a rook. For each row $r \in [n]$ we build a constraint tensor $\tau^{row,r} [X_{r \times [n]}]$, where we denote by $X_{r \times [n]} = \{X_{rc} : c \in [n]\}$ the variables to a row. The constraint tensor is defined for each coordinate $x_{r \times [n]} \in [2]^n$ by

$$\tau^{row,r} [X_{r \times [n]} = x_{r \times [n]}] = \begin{cases} 1 & \text{if } \sum_{c \in [n]} x_{rc} \leq 1 \\ 0 & \text{else} \end{cases}.$$

Vise versa, for each column $c \in [n]$ a constraint tensor

$$\tau^{col,c} [X_{[n] \times c} = x_{[n] \times c}] = \begin{cases} 1 & \text{if } \sum_{r \in [n]} x_{rc} \leq 1 \\ 0 & \text{else} \end{cases}.$$

- (ii) We introduce for each constraint a decomposition variable I of dimension $n + 1$
- For $i \in [n]$ the assignment $I = i$ shall represent that the i th atomic variable has a 1 assigned.
 - The state $I = n$ shall represent that none of the atomic variables has a 1 assigned.

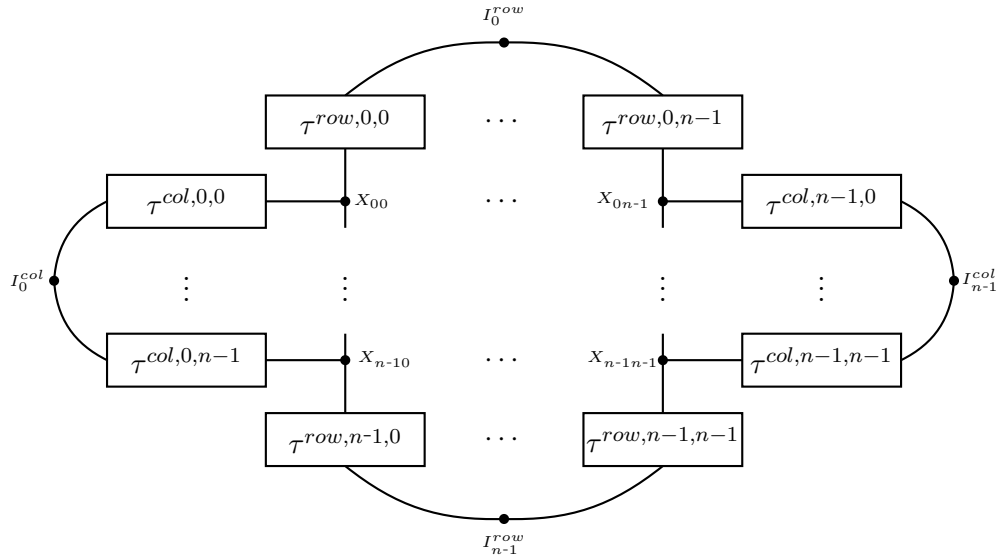
For row constraints this decomposition variable is enforced for each $c \in [n]$ by the matrices

$$\tau^{row,r,c} [X_{rc}, I_r^{row}] = \epsilon_1 [X_{rc}] \otimes \epsilon_c [I_r^{row}] + \epsilon_0 [X_{rc}] \otimes (\mathbb{I} [I_r^{row}] - \epsilon_c [I_r^{row}]).$$

We then have

$$\tau^{row,r} [X_{r \times [n]}] = \langle \{ \tau^{row,r,c} [X_{rc}, I_r^{row}] : c \in [n] \} \rangle_{[X_{r \times [n]}]}.$$

For column constraints we find analogous decomposition variables, by interchanging the r and c indices. In total we have a constraint network sketched as:



Task 2:

If there is a row or column with two rooks assigned, then there is no assignment satisfying all constraints. If not, then there are

$$(n - r)! = 1 \cdot 2 \cdot \dots \cdot (n - r)$$

possible assignments. Lets denote by E the set of the r cell tuples with a rook assigned. The total number of assignments respecting all constraints is the contraction

$$\left\langle \left(\bigcup_{(r,c) \in E} \epsilon_1 [X_{rc}] \right) \cup \left(\bigcup_{r,c \in [n]} \{ \tau^{row,r,c} [X_{rc}, I_r^{row}], \tau^{col,c,r} [X_{rc}, I_c^{col}] \} \right) \right\rangle_{[\emptyset]} .$$